

LOIC FERNANDEZ

Full-Stack Engineer — Go · Node.js · TypeScript · Python · LLM Systems

+33 6 74 31 24 37

l.frz34@proton.me

github.com/Angfr95

portfolio.ultizion.com

Engineer with a double Master's (Audit & Management) who architects and ships production systems end-to-end. Five years building distributed platforms independently — from data models and API contracts to Kubernetes, CI/CD, and LLM-powered tooling. I thrive at the intersection of engineering depth and customer impact. Fluent French (native) · English C1.

TECHNICAL SKILLS

Languages	Go, TypeScript, JavaScript, Python, SQL
Backend	Node.js, NestJS, Express, Gin (Go), FastAPI, REST, gRPC, GraphQL, Zod
LLM / AI	Anthropic API, OpenAI, OpenRouter, LangChain patterns, agentic pipelines, RAG, prompt engineering
Messaging	BullMQ (prod), NATS JetStream (prod), RabbitMQ (studying), Kafka (studying)
Databases	PostgreSQL, Redis, ClickHouse, MongoDB
Frontend	React, Vite, Tailwind CSS, Zustand, react-i18next
DevOps	Docker, Kubernetes, GitHub Actions, AWS EC2, PM2, Nginx, Prometheus, Jaeger

PROFESSIONAL EXPERIENCE

Senior Full-Stack Engineer & Independent Consultant

2021 – Present

End-to-end architecture, development, and delivery of distributed platforms and LLM-powered tooling.

Ultizion — Production E-commerce Platform

ultizion.com · 2021 – Present

Node.js · TypeScript · PostgreSQL · Redis · BullMQ · Stripe · React · AWS EC2 · Docker · GitHub Actions

- Scoped and delivered end-to-end a 14-module backend — data models, API contracts, and event-queue strategy from scratch, with no spec and no team.
- Deployed an async price-alert pipeline (Redis pub/sub + BullMQ) guaranteeing price updates delivered in under 60 min; integration-tested with Vitest across 500+ SKUs.
- Automated zero-downtime deployments on AWS EC2 via GitHub Actions + PM2 symlink swaps, achieving 99.9% uptime and eliminating service interruptions during releases.
- Shipped a multilingual frontend (FR/EN, 27 pages) with sub-2s load times — from wireframes to production, independently.

AeroPath — Aviation Learning Platform

github.com/aeropath · 2024 – Present

Go · Gin · gRPC + Protobuf · PostgreSQL · Redis · NATS JetStream · ClickHouse · Kubernetes · Prometheus · Jaeger

- Architected and deployed an event-driven Go backend (API Gateway + async Worker on NATS JetStream) with a spaced-repetition engine and real-time exam analytics.
- Optimised storage and compute costs via polyglot persistence: ClickHouse ingesting 10M+ analytics events for real-time dashboards, NATS for event scaling, Redis for caching, PostgreSQL for transactional data.
- Configured a Kubernetes cluster with HPA (3–10 API pods, 2–5 workers), StatefulSets, and Ingress TLS — sustaining load spikes during exam sessions without manual intervention.
- Instrumented end-to-end observability: distributed tracing with Jaeger, metrics with Prometheus, internal comms via gRPC + Protobuf.

FINDER — Automated Web Security Auditor

2023 – Present

Python · FastAPI · Anthropic API · OpenAI · ML (RandomForest) · Docker

- Built a security audit tool covering 30+ blackbox modules and 11+ whitebox modules — classic vectors (SQLi, SSRF, XXE, SSTI, JWT) and distributed systems (Kafka, RabbitMQ, gRPC, mTLS, service mesh).
- Integrated an LLM triage pipeline (Anthropic/OpenAI/Ollama) for intelligent vulnerability scoring: confidence thresholds, deduplication, and correlated attack-chain reports.
- Orchestrated scans via an autonomous agent: context-aware follow-up probes, adaptive thresholds, PDF/HTML output reports.

EDUCATION

Master's in Audit & Internal Control + Master's in Research & Management

2019 – 2021

University of Montpellier

Erasmus Exchange — Ireland

2018 – 2019

English C1 — full academic year abroad